The Climate Standoff

A Dynamic Game Model of Strategic Tipping in Global Climate Coordination

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Declaration

I certify that all material in this dissertation which is not my own work has been identified and that no material is included for which a degree has previously been conferred upon me. The work presented here is my own.

Signed: David Clements Date: April 2026

Abstract

Global temperatures and the frequency of extreme climate events continue to rise, yet international climate coordination remains uneven. In some regions it is accelerating; in others it has stalled. What determines which path emerges?

This paper develops a finite-horizon dynamic game among four heterogeneous blocs (US, EU, China, and a developing-economy aggregate). Each bloc chooses whether to irreversibly commit to a green transition. Adoption decisions then reshape the strategic environment along three margins: they lower transition costs, raise the political and trade cost of holding out, and generate non-excludable benefits once a global stabilisation threshold is reached. Parameters are grounded in macroeconomic and emissions data and calibrated by simulated method of moments (SMM). Coordination dynamics are analysed through marginal parameter sweeps and global sensitivity analysis.

Two distinct channels emerge. Taken one parameter at a time, patience dominates, with joint increases in US and Chinese patience producing the largest gains in coordination. When parameters move together, transition costs dominate instead, and China's cost channel is particularly decisive for global outcomes. US policy predictability anchors convergence independently of the two main channels.

A second result concerns coalition formation. Lowering adoption costs primarily benefits blocs already inclined to move, leaving first-mover incentives largely unchanged. Mechanisms that raise the cost of non-participation do the opposite. They shift the incentives of holdouts directly, which makes them more effective at triggering the initial coordination on which everything else depends.

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1 Introduction

The puzzle: why cooperation accelerates

Threshold public goods models explain coordination failure through two interlocking mechanisms: collective benefits depend on reaching a critical participation threshold, and multiple equilibria make that threshold difficult to attain. This literature gives a clear account of why cooperation stalls. It gives a much weaker account of why cooperation sometimes accelerates.

Recent climate policy illustrates the gap. The EU's Carbon Border Adjustment Mechanism and the US Inflation Reduction Act were not primarily the product of multilateral agreement. They were unilateral moves that reshaped the incentives others faced. Adopters altered the payoff structure confronting other blocs, and institutional coordination emerged as a consequence of strategic change rather than a precondition for it.

The strategic environment

The setting resembles a multilateral game of chicken. Each bloc would prefer that others absorb the upfront cost of transition, but mutual delay risks an outcome worse for everyone. At least one bloc must adopt; the analytically interesting questions are which one moves first and how that decision propagates.

The players are highly asymmetric. A small number of blocs are systemically pivotal, in the sense that their adoption decisions move global outcomes substantially on their own. Others face high transition costs against short political horizons, which weakens their incentive to move early. Still others are highly exposed to climate risk but carry too little individual weight to shift the global coordination problem through unilateral action. What this implies is that the coordination problem is not only about whether cooperation occurs. It is also about how incentives to move are distributed across blocs, and how each bloc's decision updates the expectations of the others.

Standard treaty-based models abstract from these dynamics. They take enforcement pressure, cost trajectories, and threshold progress as exogenous, and so cannot speak to how incentives evolve endogenously, when adoption is timed, or why pivotal players matter disproportionately. This paper develops a dynamic framework in which these mechanisms are endogenous, and uses it to explain both the emergence and the acceleration of climate coordination.

2 Review of Literature

Equilibrium Concepts: Nash to Quantal Response

The dominant solution concepts in the climate cooperation literature rest on a common assumption: players best-respond exactly to the strategies of others. Barrett (1994) uses stage-game Nash equilibrium and renegotiation-proof repeated-game equilibrium to characterise self-enforcing international environmental agreements. Harstad (2012) selects Markov perfect equilibrium on the grounds that strategies depending only on payoff-relevant state variables are both analytically tractable and behaviourally plausible. In Harstad's dynamic stock-pollution setting, Markov strategies generate a self-interested incentive to cut emissions: each player internalises that higher stocks trigger stronger reactions from others, which produces gradual cooperation without external enforcement.

The assumption of exact best response is not incidental to these frameworks. It is load-bearing. Harstad's Markov equilibrium does generate a cooperation gradient within the dynamic game, but the gradient rests on players who compute and commit to exact best responses to the pollution stock. That becomes a demanding assumption when the players are governments subject to domestic political friction.

Carraro and Siniscalco (1993) establish the classical fragility result: stable coalitions under internal and external stability conditions are typically small, and because participation decisions are discrete, small parameter changes induce large discontinuous shifts in equilibrium coalition size. The framework moves abruptly between near-full cooperation and near-complete defection, which limits its usefulness precisely where the empirical record is most interesting: at the margin, where small incentive changes produce cascading adoption rather than stasis. Nash provides a clean benchmark for stability, but it offers no gradient for change.

Quantal response equilibrium resolves this by replacing exact best response with probabilistic choice (McKelvey and Palfrey, 1995). Players choose better options more often, but not always. Choice probabilities increase in expected payoffs, scaled by the rationality parameter λ. A marginal increase in adoption costs therefore reduces adoption probability incrementally rather than flipping participation off entirely. Goeree, Holt, and Palfrey (2003) show that QRE tracks observed behaviour in several laboratory strategic environments.

Stochastic choice in QRE also has direct political-economy content. Putnam (1988) formalises why governments cannot behave as frictionless unitary optimisers: international commitments must remain feasible in the domestic ratification game. Read through that lens, QRE's choice noise admits a structural interpretation as bureaucratic and ratification friction rather than as purely behavioural error, which suits an environment in which the players are governments rather than individuals.

QRE ensures that smooth incentive changes produce smooth behavioural responses. It does not explain when those responses accumulate into self-reinforcing adoption cascades.

Strategic Tipping and Diffusion Dynamics

The physical and technological foundations of tipping are reasonably well understood. Way et al. (2022) show that endogenous cost declines driven by fast learning curves for renewables can shift mitigation pathways more sharply than most integrated assessment models assume. That finding sits comfortably in the directed technical change literature, where Acemoglu et al. (2012) establish that innovation incentives respond to deployment scale in ways that can generate self-reinforcing cost trajectories. Farmer et al. (2019) introduce sensitive intervention points in socioeconomic systems and argue that small, well-targeted interventions can exploit these amplification mechanisms to trigger cascading low-carbon transitions. Otto et al. (2020) identify specific social subsystems where positive feedback could rapidly compress emissions trajectories. Building on that, Sharpe and Lenton (2021) frame upward-scaling tipping cascades, in which activating one socio-technical tipping point raises the probability of triggering the next. Milkoreit et al. (2018) formalise the social tipping point concept across interdisciplinary contexts, identifying belief shifts and perceived feasibility as distinct triggers alongside cost and technology channels.

These contributions capture dynamics observed in real energy transitions, but they share a structural limitation: adoption decisions are embedded in aggregate diffusion mechanisms, systems feedback frameworks, or reduced-form cost trajectories, rather than arising from explicit strategic interaction between heterogeneous countries. This matters because the mechanism becomes difficult to identify. Comin and Hobijn (2010) document precisely this puzzle in a long historical panel of technology diffusions: structurally similar countries adopt at systematically different speeds, and institutional and political factors explain residual variation that cost differentials alone cannot. Without an explicit strategic foundation, cascades generated by cost feedbacks look much like cascades generated by expectation shifts or by politically mediated peer effects, yet each carries different policy implications.

A second problem compounds the first. Tipping behaviour in formally specified models is often sensitive to chosen parameter values, which makes it difficult to distinguish structurally robust dynamics from calibration artefacts. A finding that holds only at a particular cost trajectory or rationality parameter is not a finding about tipping. It is a finding about one point in the parameter space.

Taken together, these gaps motivate a framework that endogenises tipping through asymmetric strategic interaction, with a disciplined method for establishing which dynamics survive joint parameter perturbation.

Gaps and Contribution

Two gaps run through the literature reviewed above. The first is structural. No existing framework jointly endogenises the three forces the empirical record suggests are co-moving: transition costs that decline as the coalition grows, political pressure that escalates on holdouts as the coalition closes around them, and the asymmetric influence of pivotal blocs whose individual moves shift others' incentives. Coalition models such as Barrett (1994) and Nordhaus (2015) characterise the participation set at a static club equilibrium. Nordhaus endogenises the environment through trade penalties on non-members, but the equilibrium he solves is still a single-shot membership determination rather than a trajectory through time. Diffusion models such as Grubb, Hourcade, and Neuhoff (2014) capture aggregate cost trajectories but not the strategic identity of who is adopting and in what order. Neither tradition explains how early adoption by anchor players reshapes the incentives of holdouts over time.

The second gap is methodological. Kolstad (2007) documents that quantitative results in the IEA literature shift substantially across plausible cost and benefit assumptions, yet few studies distinguish the cross-bloc structure pinned down by observable data from the latent cardinal strength of the payoff channels. Most approaches calibrate to a single benchmark or rely on ad hoc sensitivity checks, which leaves the reader unable to tell whether a finding reflects a structural property of the game or a feature of one point in the parameter space.

This paper addresses both gaps. The core framework is a finite-horizon Markov game with heterogeneous blocs and irreversible adoption, solved under QRE. This delivers smooth comparative statics without the brittleness that Nash-based coalition models inherit from discrete participation decisions.

Calibration is separated into two layers. The ordinal layer is identified from macroeconomic and emissions data. The cardinal layer is recovered by simulated method of moments (McFadden, 1989), matching observable features of the contemporary climate transition: EU leadership, strategic free-riding by major emerging economies, and the delayed onset of self-reinforcing adoption. Together these produce a single fitted baseline anchored to the 2025 strategic environment.

Marginal sweeps identify which channels matter in isolation. Global sensitivity analysis (Saltelli et al., 2008) then stress-tests those rankings by jointly perturbing the full parameter space, which separates levers that are locally powerful from those that remain influential once the wider strategic environment moves with them.

3 Methodology

3.1 The Model

Players and Structure

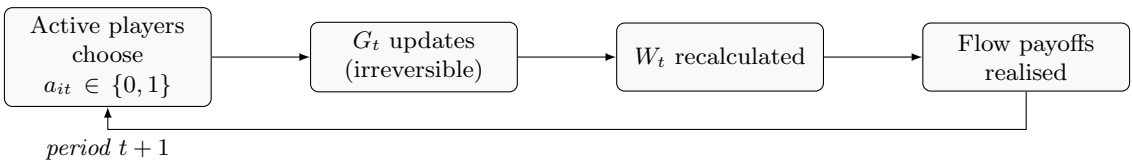
$$i \in \{\text{US, EU, CN, RoW}\}$$

Each player represents a geopolitical bloc with distinct economic structure, emissions profile, and strategic incentives. The US, EU, and China dominate global emissions, renewable manufacturing, and geopolitical weight; the Rest of World aggregates developing and emerging economies whose individual influence is limited but whose collective climate vulnerability is high. The blocs with most power to act are least directly threatened by delay; the bloc most threatened has least power to act alone.

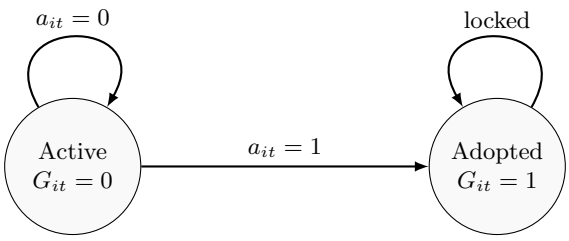
Time is discrete and finite, $t = 1, \ldots, T$, with $T = 10$ five-year periods representing the 50-year mitigation window beyond which further delay locks in irreversible physical damages (IPCC AR6; Lenton et al., 2019). In each period, active players simultaneously choose $a_{it} \in \{0, 1\}$, with adoption status evolving as

$$G_{i,t+1} = G_{it} + (1 - G_{it})\, a_{it}, \qquad G_{i1} = 0.$$

Once $G_{it} = 1$ the player's status is permanent, reflecting the sunk character of grid and capital investments. The active set is $\mathcal{A}_t = \{i : G_{it} = 0\}$. Each bloc carries a normalised influence weight $w_i > 0$, $\sum_i w_i = 1$, so that cumulative weighted adoption is $W_t \equiv \sum_i w_i G_{it}$ and the payoff-relevant state is $s_t = (t, G_t)$ with $G_t = (G_{1t}, \ldots, G_{4t})$. Backward induction tracks the full vector G_t rather than the scalar W_t because payoffs are player-specific.



Game flow diagram. Each period, active players simultaneously choose whether to adopt. The adoption vector updates, weighted coalition mass W_t is recalculated, and flow payoffs are realised before the game advances to the next period.



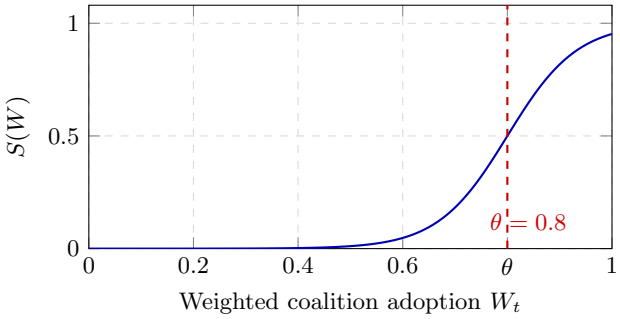
Lock-in and lock-out dynamics implied by the law of motion: once a bloc adopts, its status is permanent.

Coordination Threshold and Stabilisation

Collective benefits are captured by a smooth sigmoid centred at the coordination threshold θ:

$$S(W) = \frac{1}{1 + \exp(-\eta(W - \theta))}$$

where $\eta > 0$ controls sharpness. $S(W)$ is flat for $W \ll \theta$, steep in a band around θ, and approaches a step function as $\eta \to \infty$. Both θ and η are varied in the sensitivity analysis.



Stabilisation benefit as a function of weighted coalition adoption. The sigmoid $S(W)$ activates sharply as W_t crosses the coordination threshold $\theta = 0.8$ under the baseline steepness $\eta = 15$; the dashed line marks where collective action tips from partial to near-full stabilisation.

Flow Payoffs

All flow payoffs in period t are evaluated at W_t, the cumulative adoption state at the *start* of the period, before current actions are realised. Under simultaneous choice, no player observes others' period-t decisions when choosing. Adopters incur the one-off transition cost in the period of adoption; any learning spillover from their own action enters only future periods via W_{t+1}.

Transition Costs:

$$c_i(W_t) = c_i^0 \, (1 - \gamma \, W_t)$$

One-off cost paid in the period of adoption. Baseline burden c_i^0 declines with coalition size at learning rate γ.

Political Pressure:

$$p_i \cdot W_t$$

Per-period cost borne by active holdouts, linear in coalition size. p_i is the bloc's exposure to trade restrictions and diplomatic isolation.

Climate Damages:

$$d_i(t, W_t) = d_i^0 \cdot (1 + \kappa t) \cdot (1 - S(W_t))$$

Non-excludable. Baseline vulnerability d_i^0 escalates at rate κ and attenuates as coordination approaches the threshold.

Stabilisation Benefit:

$$b \cdot S(W_t)$$

Non-excludable collective payoff received by all players as W_t crosses θ. b is the aggregate weight on avoided global damages, recovered by SMM.

Combined Flow Payoffs:

The state-contingent component received by all players regardless of action:

$$r_i(t, W_t) = -d_i(t, W_t) + b \cdot S(W_t)$$

For active players ($i \in \mathcal{A}_t$), the action-specific payoffs are:

$$r_i^A(t, W_t) = -c_i(W_t) + r_i(t, W_t)$$
$$r_i^D(t, W_t) = -p_i\, W_t + r_i(t, W_t)$$

For already-adopted players ($i \notin \mathcal{A}_t$), the transition cost has been paid in full and their period payoff is simply $r_i(t, W_t)$.

Discount Factors:

Each player carries a discount factor $\delta_i \in (0, 1)$ applied to period-$t + 1$ payoffs. Bloc-specific values are set in the Calibration section.

Composite Spillover Channel:

The cost-learning and political-pressure mechanisms both load on the same state variable W_t: falling costs pull potential adopters in, rising pressure pushes holdouts in. No moment constructed from adoption trajectories can separate them, so the two are collapsed into a single spillover channel with mixing parameter $\phi \in [0, 1]$:

$$c_i(W_t) = c_i^0 - \tilde{c}_i\,(1 - \phi)\,\gamma\, W_t$$
$$\text{pressure}_i(W_t) = \phi\, p_i\, W_t$$

ϕ is held at 0.5 during estimation and swept separately in Appendix B. The cost-learning side is the *carrot*; the pressure side is the *stick*.

Recursive Value Structure

Each bloc is forward-looking. When deciding whether to adopt or wait in a given period, it weighs the immediate payoff of each choice against the discounted value of all future periods, taking others' likely behaviour into account. The two action-specific values are written as Q_i^A (adopt) and Q_i^D (delay), each adding the current flow payoff to the discounted expected continuation value:

$$Q_i^A(s_t) = r_i^A(t, W_t) + \delta_i \, \mathbb{E}[V_i(s_{t+1}) \mid a_{it} = 1]$$
$$Q_i^D(s_t) = r_i^D(t, W_t) + \delta_i \, \mathbb{E}[V_i(s_{t+1}) \mid a_{it} = 0]$$

The conditioning on a_{it} matters: player i's own action shifts W_{t+1} by w_i if they adopt, altering the distribution over future states and the expected continuation value of each choice. The recursion ends with the terminal condition $V_i(T+1, \cdot) = 0$.

The continuation value is the log-sum-exp form arising from i.i.d. Type I extreme-value (Gumbel) shocks on each action with scale $1/\lambda$ (McFadden, 1973):

$$V_i(s_t) = \frac{1}{\lambda} \log\Big(e^{\lambda Q_i^A} + e^{\lambda Q_i^D}\Big)$$

As $\lambda \to \infty$, V_i collapses to $\max\{Q_i^A, Q_i^D\}$, recovering the standard Bellman equation.

For already-adopted players ($i \notin \mathcal{A}_t$):

$$V_i(s_t) = r_i(t, W_t) + \delta_i \, \mathbb{E}[V_i(s_{t+1})]$$

Future utility continues to depend on the W trajectory driven by the remaining active players.

Equilibrium

Equilibrium adoption probabilities are defined over active players and follow the logit QRE formula. For $i \in \mathcal{A}_t$:

$$\sigma_i(s_t) = \frac{\exp\{\lambda Q_i^A\}}{\exp\{\lambda Q_i^A\} + \exp\{\lambda Q_i^D\}}$$

For $i \notin \mathcal{A}_t$, $\sigma_i(s_t) = 0$ by the irreversibility constraint. Equilibrium is a fixed point in the mapping from beliefs over opponents' actions into optimal logit response probabilities:

$$\sigma^* = \mathrm{QRE}(Q(\sigma^*))$$

No player wishes to revise their strategy given others' strategies, and others' strategies are consistent with those beliefs. The game is solved by backward induction from period T. At each state, fixed-point iteration finds the self-confirming probability vector.

Solution

The game is solved by backward induction from the terminal condition $V_i(T+1, \cdot) = 0$. Irreversibility bounds the adoption state space to $2^4 = 16$ configurations per period, so exhaustive enumeration is trivial. For a given action profile, the transition probability factors as independent Bernoulli trials,

$$\Pr(G_{t+1} \mid G_t, \sigma) = \prod_{j \in \mathcal{A}_t} \sigma_j(s_t)^{a_{jt}} \left(1 - \sigma_j(s_t)\right)^{1-a_{jt}},$$

and expected continuation values are taken as $\mathbb{E}[V_i(s_{t+1}) \mid a_{it}] = \sum_{G_{t+1}} \Pr(G_{t+1} \mid G_t, \sigma, a_{it}) \cdot V_i(t+1, G_{t+1})$.

3.2 Data and Calibration

Implementing the model requires assigning numerical values to all parameters. Some can be disciplined directly by observable data. Others cannot. The calibration strategy addresses this by separating the problem into two layers, each with a distinct identification logic.

The first layer fixes ordinal structure: the relative positioning of blocs within each parameter channel. Cross-bloc rankings in transition costs, climate damages, and systemic influence can be inferred from data. The EU's position is identified from carbon intensity, trade openness, and deployment history, so that the SMM moments then target the magnitude of the EU to peer gap rather than its existence. World Bank (2026) development indicators, national emissions inventories, and bilateral trade statistics provide the evidence base. This layer is held fixed throughout, because perturbing it would mean contradicting the observable record.

The second layer recovers cardinal scaling. The relative importance governments place on transition costs, adoption spillovers, climate damages, and coordination benefits is not observable from macroeconomic data (Nordhaus, 2007) and is estimated using simulated method of moments (McFadden, 1989).

Ordinal structure is data-identified. Cardinal scaling is structurally estimated. The sensitivity analysis is conducted around this fitted baseline, which is what allows the distinction between locally powerful levers and structurally robust ones to carry empirical weight.

3.2.1 Layer 1: Ordinal Structure

Each per-player parameter is constructed from bloc-level World Bank (2026) data through a structural transformation anchored to the EU as the reference bloc, so that a single scaling α sets the EU's level and the other blocs follow data-implied ratios.

The EU is used as the numeraire because its long continuous record of climate legislation and ETS operation makes its bloc-level values the best-anchored data point. This is a normalisation convention only. The EU/US gap that appears later as a calibration moment is a moment about magnitude rather than about which bloc is the anchor, so no circularity is introduced.

Transition costs

are derived from the carbon intensity of GDP, adjusted for development. Raw intensity is misleading for less developed economies, where low values often reflect limited industrialisation rather than clean infrastructure. A development-gated interpolation blends each bloc's observed intensity with a global baseline, weighted by GDP per capita relative to the richest bloc. An affordability penalty (elasticity $\epsilon = 0.3$) captures the higher cost of capital for renewable deployment in developing economies (Ameli et al., 2021). Together these adjustments ensure the ordinal cost structure reflects genuine transition burdens.

Climate damages

are proxied by the share of agriculture, forestry, and fishing in GDP, capturing structural economic exposure to climate risk. Burke et al. (2015) document that agricultural productivity losses account for the largest share of measurable climate damages across income levels. The proxy is preferred over composite indices like ND-GAIN for its direct observability and cross-bloc consistency.

Political pressure

is proxied by trade openness (trade as a share of GDP). A more trade-dependent economy has more to lose from non-cooperation: carbon border adjustments, climate-linked conditionality, and diplomatic friction all arrive through trade relationships.

Influence weights

combine emissions share and GDP share in a 0.30/0.70 weighting. Emissions share captures necessity for any effective coalition, while GDP share captures the economic channels through which adoption influence operates. The 70/30 weighting is varied in the robustness checks reported in the appendix.

Special Note on Player Discount Rates:

Discount factors δ_i reflect the effective length of each bloc's credible planning horizon: how far into the future its institutions act as if they weight future payoffs (Nordhaus, 2007). They are deliberately descriptive rather than prescriptive.

The EU receives $\delta_{EU} = 0.85$ (implied annual rate $\approx 3.3\%$), reflecting the legal bindingness of the European Climate Law and over two decades of continuous ETS operation. China is assigned $\delta_{CN} = 0.80$ ($\approx 4.6\%$). Centralised governance supports continuity through the five-year plan system, but growth constraints shorten the effective horizon. The United States receives $\delta_{US} = 0.75$ ($\approx 5.9\%$), reflecting the historically shorter effective planning horizons produced by frequent electoral cycles, divided government, and the constitutional separation of powers. The Paris withdrawal and re-entry episode is read primarily through the rationality channel λ_{US} rather than through δ_{US}, since it reflects domestic ratification volatility rather than a structural compression of the planning window. The Rest of the World is assigned $\delta_{RoW} = 0.70$ ($\approx 7.4\%$ annually), consistent with the GDP-weighted sovereign borrowing costs of major developing economies including India, Indonesia, Brazil, and Mexico (World Bank, 2026; Ameli et al., 2021). The ordering EU > China > US > RoW holds across all specifications, and the blocs with the greatest patience are not those with the greatest climate vulnerability. This is a central structural tension of the model.

These values are reasoned from observable institutional features rather than estimated, so the specific ordering is itself a narrative choice. Appendix D reports how the headline patience result behaves under five alternative orderings, including a US/CN flip and a fully equalised vector.

Table 1: Bloc-level discount factors. Per-period values apply to a five-year model period; the implied annual rate is $\delta_i^{-1/5} - 1$, the constant annual yield that compounds to the per-period factor over five years.

Bloc	**Per-period δ_i**	**Implied annual rate**
European Union	0.85	3.3%
China	0.80	4.6%
United States	0.75	5.9%
Rest of World	0.70	7.4%

Global Parameters:

Global parameters describe the structure of the game rather than the characteristics of individual players. They cannot be identified through cross-bloc comparisons, so each is anchored to a baseline drawn from published estimates in the climate economics and game theory literature. These baselines are reference points, not fixed values. All global parameters are varied systematically across the analysis.

Coordination Threshold:

The coordination threshold ($\theta = 0.80$) defines the weighted participation level at which stabilisation benefits become substantial. The Paris Agreement's 55% emissions threshold governs entry into force, but entry into force is not effective mitigation. Net-zero pathways in IEA (2021) and IPCC 1.5°C scenarios consistently require near-universal engagement among major emitters. The four blocs modelled here account for the substantial majority of global emissions, so a threshold of 0.80 reflects the near-universal requirement those pathways imply. As with other global parameters, θ is varied in the sensitivity analysis.

Sigmoid Steepness:

The sigmoid steepness parameter η determines how sharply stabilisation benefits concentrate around the threshold. High values produce tipping-point dynamics in which benefits activate rapidly once θ is crossed. Lower values generate a more gradual accumulation of benefits across adoption levels. The baseline ($\eta = 15$) implies a steep transition, with $S(W)$ rising from approximately 0.06 to 0.94 within a ± 0.2 band around θ. It is treated as an experimental lever.

QRE Rationality:

The precision parameter λ determines how sharply each bloc's strategy responds to payoff differences. Experimental estimates typically range from 0.5 to 10 depending on context (McKelvey and Palfrey, 1995; Goeree, Holt, and Palfrey, 2003). The baseline sets $\lambda = 1.5$ as a common global benchmark, reflecting the persistent noise in international climate negotiations from domestic political constraints, electoral volatility, and bureaucratic friction. Bloc-specific values are introduced as experimental levers in the analysis to capture institutional heterogeneity that a single global parameter cannot.

Technology Learning Rate:

The learning rate ($\gamma = 0.25$) captures cost reductions driven by cumulative adoption through technology spillovers, supply chain development, and learning-by-doing. Empirical estimates suggest declines of 20% to 25% per doubling of capacity for solar PV and 10% to 15% for wind (Rubin et al., 2015; IRENA, 2023). The chosen value aligns with the upper end of these estimates, reflecting that "adoption" here represents a broad technological transition rather than a single technology.

Damage Escalation:

The damage escalation rate $\kappa = 0.05$ sets how sharply climate damages compound when mitigation is delayed. Over the model's 50-year horizon, this implies that baseline damages rise by roughly 50% under sustained non-cooperation. That figure is consistent with IPCC AR6 estimates, which place the cost of delayed mitigation at 1% to 2% of GDP per decade.

3.2.2 Layer 2: Simulated Method of Moments (SMM) Calibration

SMM recovers four cardinal scaling parameters by minimising the weighted squared distance between six model-implied moments and a set of benchmark targets representative of the 2025 climate-policy environment. The mixing parameter ϕ is held fixed at 0.5 and varied separately in the sensitivity analysis reported in Appendix B. That analysis shows the recovered cardinal vector is stable across the identifiable range of ϕ: the four estimates do not move materially when the spillover composition is reweighted.

Each α has a direct economic reading:

- α_c: weight on upfront transition costs.
- α_{spill}: responsiveness to coalition growth (carrot and stick combined).
- α_d: weight on present-day climate damages.
- α_b: weight on the collective payoff from stabilisation.

Targeted Moments:

Six benchmark targets discipline the calibration. Each is a stylised anchor reflecting expert judgement about the 2025 strategic environment, because no observational estimates exist for what are fundamentally forward-looking adoption decisions. Three *static* benchmarks at period one (where $W_0 = 0$) identify the baseline cost, damage, and benefit weights. Three *dynamic* benchmarks at period two or across timing identify the spillover channel.

- **M1: EU versus US adoption gap, period one** (0.25). Captures persistent EU leadership in climate policy and deployment relative to a more fragmented US stance (Wurzel et al., 2019; Falkner, 2016).
- **M2: US period-one adoption probability** (0.10). Historically limited early US adoption reflects political fragmentation and free-riding incentives (Nordhaus, 2015; World Bank, 2026).
- **M3: China period-one adoption probability** (0.10). Structural cost constraints and development priorities limit early commitment (IEA, 2025; WRI, 2025; Hallegatte et al., 2016).
- **M4: China period-two / period-one adoption ratio** (2.0). Early coalition formation is well documented to accelerate subsequent adoption incentives through cost and strategic spillovers (Farmer et al., 2019; Way et al., 2022; Acemoglu et al., 2012). Because $W_0 = 0$ carries no spillover signal, this moment isolates α_{spill}.
- **M5: US expected period-two adoption probability** (0.15). Under domestic ratification constraints, US adoption adjusts gradually rather than converging immediately (Putnam, 1988; Aklin and Urpelainen, 2013).
- **M6: Mean coordination timing conditional on success** (5.0 periods). Consistent with widely cited net-zero pathways requiring coordination by mid-century (IEA, 2021; IPCC, 2021).

These benchmarks are deliberately stylised, and the policy hierarchy reported in Section 6 is robust to reasonable variation in them.

Table 2: SMM Moment Targets and Model-Implied Values at Calibrated Parameters. Moments M1–M3 are evaluated at $W_0 = 0$ and identify the static parameters; M4–M5 capture period-2 acceleration and identify the spillover channel; M6 anchors aggregate timing.

Moment	**Description**	**Target**	**Model-implied**
M1	EU–US adoption gap, $t = 1$ ($\sigma_{EU,1} - \sigma_{US,1}$)	0.250	0.215
M2	US period-1 adoption probability ($\sigma_{US,1}$)	0.100	0.069
M3	China period-1 adoption probability ($\sigma_{CN,1}$)	0.100	0.128
M4	China period-2 / period-1 adoption ratio	2.000	2.113
M5	US expected period-2 adoption probability	0.150	0.147
M6	Mean coordination timing | success	5.000	4.907

Identification Strategy:

The spillover channel only switches on once a bloc has adopted. In period one, no bloc has yet adopted, so neither cost learning nor pressure has activated. Period-one adoption therefore reflects only the baseline cost burden, climate damages, and coordination benefits. The first three moments are all measured in period one, so they identify α_c, α_d, and α_b cleanly, without any interference from the spillover channel.

By period two the state has moved. The EU usually adopts in period one under the calibrated baseline, which is enough to start the spillover channel working on the remaining blocs. Moments M4 and M5 capture exactly this activation: the first records how much China's adoption probability accelerates between periods, and the second records the expected period-two adoption probability of the US. Because α_c, α_d, and α_b are already pinned down by the period-one moments, M4 and M5 carry identifying information for the spillover scaling parameter α_{spill} specifically. M6 anchors the overall speed at which coordination tips once the coalition starts to form.

The system is over-identified by two restrictions: six moments, four parameters. A perfect fit is therefore unavailable by construction, and the estimator must find a single vector that satisfies all six moments simultaneously. The SMM objective surface is not smooth, so Nelder-Mead (Nelder and Mead, 1965) is preferred to gradient-based alternatives. The optimisation is run from multiple starting points, and all four parameters recover at interior solutions.

Table 3: Estimated and Fixed Parameter Vector

Symbol	**Description**	**Value**	**Status**
SMM-estimated (Nelder-Mead, best of 4 starts)			
α_c	Baseline transition cost	3.5844	Estimated
α_{spill}	Adoption spillover scaling	8.3153	Estimated
α_d	Climate damage scaling	0.6312	Estimated
α_b	Coordination benefit scaling	0.9496	Estimated
Fixed structural			
ϕ	Spillover mixing (cost-learning vs pressure)	0.50	Fixed
λ	QRE rationality (homogeneous)	1.54	Fixed
δ_{US}	US discount factor	0.75	Fixed
δ_{EU}	EU discount factor	0.85	Fixed
δ_{CN}	China discount factor	0.80	Fixed
δ_{RoW}	Rest-of-World discount factor	0.70	Fixed
η	Sigmoid sharpness	15.0	Fixed
θ	Coalition threshold	0.80	Fixed
γ	Learning spillover rate	0.25	Fixed
κ	Damage accumulation speed	0.05	Fixed

Note: SMM minimises weighted squared distance between model-implied and empirical moments M1–M6. ϕ controls the internal split between cost learning and pressure within the spillover channel; it is held fixed at 0.5 during estimation and varied in sensitivity analysis.

3.2.3 Verification and Stability

Five verification checks are applied.

1. Restarting the optimiser produces zero parameter drift, which indicates a genuine local minimum.

2. Across five random seeds, coordination success rates have standard deviation below 0.02.

3. All five empirical ordinal rankings are preserved.

4. The Andrews, Gentzkow and Shapiro sensitivity matrix

$$\Lambda = -(G'WG)^{-1}G'W,$$

   computed via central-difference Jacobian $G = \partial m/\partial\theta$, reports the percentage change in each $\hat{\theta}_j$ induced by a one-percent change in moment m_k (Andrews, Gentzkow and Shapiro, 2017). The full M6 column is structurally elevated ($\alpha_{\text{spill}} = 12.93$, $\alpha_b = 7.80$, $\alpha_d = 4.72$, $\alpha_c = 2.46$). M6 is the aggregate timing moment, so shifting it mechanically redistributes across all four parameters that jointly govern when coordination tips. Outside the M6 column, every elasticity is below 2.0.

5. The Jacobian condition number is 202, which indicates that the moments carry independent information about each parameter (Hansen, 1982; Belsley et al., 1980).

Table 4: SMM Estimation Verification and Stability Checks

Check	**Description**	**Result**	**Verdict**
1. Optimiser restart	Max parameter shift from restart	0.0000	**PASS**
2. MC seed stability	Std. dev. of success rate across 5 seeds	<0.02	**PASS**
3. Ordinal rankings	Empirical ordinal rankings preserved	All 5	**PASS**
4. Moment sensitivity	Max AGS elasticity $\lvert\Lambda_{jk}\rvert$; $\alpha_{\text{spill}} \leftarrow$ M6 large	12.93	**NOTE**
5. Jacobian condition	Condition number of moment Jacobian	202 (<1000)	**PASS**

4 Experimental Design

4.1 What This Model is For

The model is not a forecasting tool. Its purpose is to identify which model inputs correspond to real policy instruments capable of moving the coordination outcome substantially, and which do not. Both experiments below take the calibrated baseline as a starting point.

4.2 Marginal Parameter Sweeps

The first experiment varies one parameter at a time, holding all others at their calibrated values, re-solves the game, and records how coordination success responds. Each parameter corresponds to a recognisable policy instrument: discount factors capture institutional credibility and how far governments plan ahead; cost multipliers capture transition subsidies, concessional finance, and technology transfer; pressure multipliers capture trade conditionality and carbon border adjustments; the learning rate captures how much R&D and deployment investment lower adoption costs; the coordination threshold captures the institutional design of multilateral commitments.

Each parameter is varied in percentage-point increments from its baseline value (e.g. a 10% sweep on $\delta = 0.80$ means $\delta = 0.88$, i.e. $\delta \times 1.10$, not $\delta + 0.10$). Each response curve is summarised by two crossing points: the percentage change from baseline needed to raise coordination success above 90% and 95% respectively. These give a single number per channel: the size of policy intervention required to reach a target probability of coordination. Crossings that would require values outside their economically plausible range (discount factors above 0.95, cost multipliers below zero) are recorded as absent rather than extrapolated.

4.3 Global Sensitivity Analysis

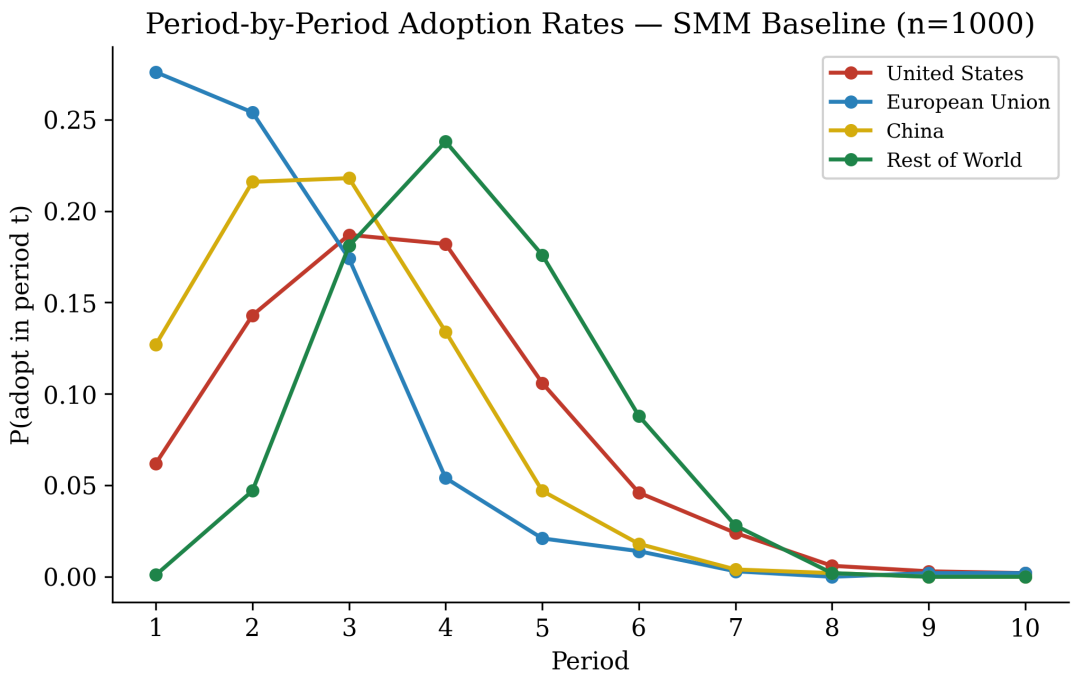
Where the sweeps test one instrument at a time, the GSA asks which inputs explain most of the variation in coordination outcomes when many parameters move at once. One thousand parameter vectors are drawn from independent uniform distributions and the game is solved at each one; the sample size balances computational feasibility against coverage of the parameter space. The four SMM-estimated cardinal weights are held fixed, so the GSA tests structural and behavioural inputs rather than payoff weights. All other inputs vary, including discount factors in their plausible ranges ($\delta_{US} \in (0.65, 0.88)$, $\delta_{EU} \in (0.75, 0.90)$, $\delta_{CN} \in (0.68, 0.88)$, $\delta_{RoW} \in (0.55, 0.78)$), cost and pressure multipliers (preserving the Layer 1 ordering across blocs), the learning rate, sigmoid steepness, bloc-level rationality, damage escalation, and the spillover mixing parameter ϕ. Full ranges are in the appendix.

The relationship between each input and each outcome is measured by Spearman rank correlation, which records how consistently a parameter and an outcome move in the same direction, regardless of whether the relationship is proportional (Iman and Conover, 1987). A parameter that ranks strongly in both the sweeps and the GSA is a structurally robust policy lever; one that ranks strongly in only one is contingent on the rest of the environment staying close to the baseline.

5 Results

5.1 Model Baselines

At the fitted calibration, coordination succeeds in 75.3% of simulated paths. The mean coordination timing conditional on success is 3.94 periods and the mean final coalition weight is 0.77. Period-one adoption probabilities are 26.8% (EU), 12.1% (CN), 6.5% (US), and 0.1% (RoW); adoption rates decline sharply after period two.



Period-by-period adoption rates $\sigma_{i,t}$ for each bloc across the ten-period horizon at the SMM-calibrated baseline. The EU peaks in period one, China and the United States in period two, and the Rest of World late around period four; rates fall toward zero by period eight as the window for successful coordination closes.

Table 5: Baseline Model Diagnostics at SMM-calibrated parameters (n = 1,000 Monte Carlo draws). Success defined as $W_T \geq \theta$.

Statistic	**Expression**	**Value**
Coordination success rate	$P(W_T \geq \theta)$	0.753
Mean final weighted adoption W_T	$\mathbb{E}[W_T]$	0.7657
Mean coordination period | success	$\mathbb{E}[\tau \mid W_\tau \geq \theta]$	3.94
Period-1 adoption prob. — US	$\sigma_{US,1}$	0.0695
Period-1 adoption prob. — EU	$\sigma_{EU,1}$	0.2848
Period-1 adoption prob. — CN	$\sigma_{CN,1}$	0.1281
Period-1 adoption prob. — RoW	$\sigma_{RoW,1}$	0.0010
First-mover freq. — US	$P(\tau_{US} = 1)$	0.0653
First-mover freq. — EU	$P(\tau_{EU} = 1)$	0.2680
First-mover freq. — CN	$P(\tau_{CN} = 1)$	0.1207
First-mover freq. — RoW	$P(\tau_{RoW} = 1)$	0.0007

5.2 Policy Levers and Structural Parameters

Results partition the parameter space into two groups. *Policy levers* are parameters with direct real-world analogues that can be influenced by policy action: discount factors δ_i (institutional credibility and effective planning horizons), cost multipliers Cost_i (subsidies, concessional finance, procurement), pressure multipliers Pressure_i (trade conditionality and diplomatic exposure), the learning rate γ (R&D and deployment-driven cost reductions), and the rationality parameters λ_i (strategic coherence of national institutions). *Structural parameters* describe the strategic landscape within which those levers act and are not directly controllable: the coordination threshold θ, the sigmoid steepness η, the spillover composition ϕ, and the damage escalation rate κ. The first group answers what can be changed; the second conditions the effectiveness of that change. The marginal sweeps (Section 6.2) and GSA (Section 6.3) both report findings under this split. Exact percentages are specific to the fitted calibration and should not be read as precise policy targets; what the simulation identifies robustly is the hierarchy across parameters.

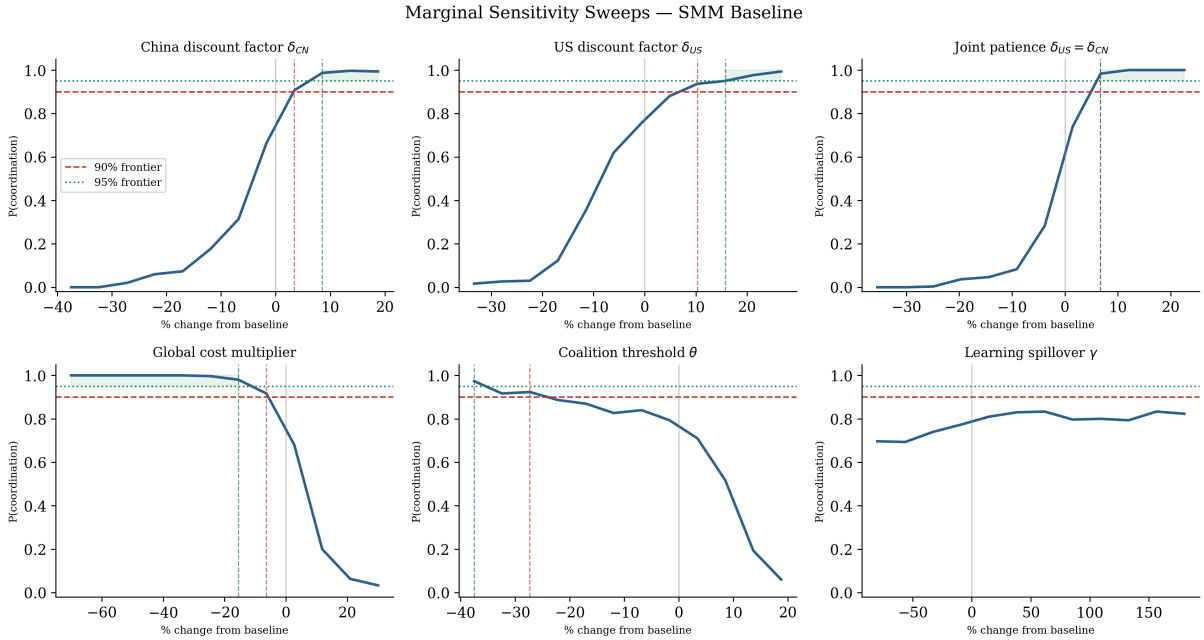
5.3 Marginal Sweep Findings

Policy levers.

Among patience channels, δ_{CN} crosses the 90% frontier at +3.4% and 95% at +8.5%; δ_{EU} at +7.5% and +11.8%; δ_{US} at +10.3% and +15.8%; δ_{RoW} at +23.1% with no 95% crossing within the swept range. The joint US/China patience sweep crosses both frontiers at +6.75%; Appendix D reports a joint crossing at +5.2% under a US/CN discount flip, while the binding individual lever switches with the ordering. Global transition cost crosses 90% at −6.4% and 95% at −15.5%; Cost_{CN} at −15.5% and −26.4%; Cost_{US} at −26.4% and −37.3%; Cost_{RoW} at −43.6% and −52.7%. US pressure crosses 90% at +58.2% and 95% at +161.8%; global pressure at +58.2% and +92.7%; CN pressure does not cross. λ_{US} crosses 90% at +213%; γ and the remaining λ channels do not cross within the swept range.

Structural parameters.

θ crosses the 90% frontier at −27.3% and 95% at −37.5%. κ crosses at +58.2% and +92.8%. η does not cross within the swept range. ϕ is held fixed at 0.5 in the marginal sweep; its effect is reported in the GSA, where it is varied across $[0, 1]$.



Key policy channel sweeps: coordination success rate vs. % deviation from SMM baseline. Dashed red = 90% frontier; dotted teal = 95% frontier.

Table 6: Marginal Sweep Threshold Crossings (SMM Baseline)

Channel	**Direction**	**90% crossing (% Δ)**	**95% crossing (% Δ)**
θ (coalition threshold)	Dec.	-27.28%	-37.50%
δ_{US}	Inc.	+10.31%	+15.76%
δ_{CN}	Inc.	+3.41%	+8.52%
$\delta_{US} = \delta_{CN}$ (joint)	Inc.	+6.75%	+6.75%
δ_{EU}	Inc.	+7.48%	+11.76%
δ_{RoW}	Inc.	+23.08%	—
Global cost multiplier	Dec.	-6.36%	-15.45%
CN cost multiplier	Dec.	-15.45%	-26.36%
US cost multiplier	Dec.	-26.36%	-37.27%
RoW cost multiplier	Dec.	-43.64%	-52.73%
γ (learning spillover)	Inc.	—	—
κ (damage accumulation)	Inc.	+58.20%	+92.80%
η (sigmoid sharpness)	Inc.	—	—
CN pressure multiplier	Inc.	—	—
US pressure multiplier	Inc.	+58.18%	+161.82%
Global pressure multiplier	Inc.	+58.18%	+92.73%
λ_{US} (QRE rationality)	Dec.	+212.87%	+437.19%
λ_{CN} (QRE rationality)	Dec.	—	—
λ_{EU} (QRE rationality)	Dec.	—	—
λ_{RoW} (QRE rationality)	Dec.	—	—

Note: Crossing values expressed as % change from the SMM-calibrated baseline. Positive values indicate the parameter must increase from baseline; negative values indicate it must decrease. — indicates crossing absent within sweep range.

5.4 GSA Findings

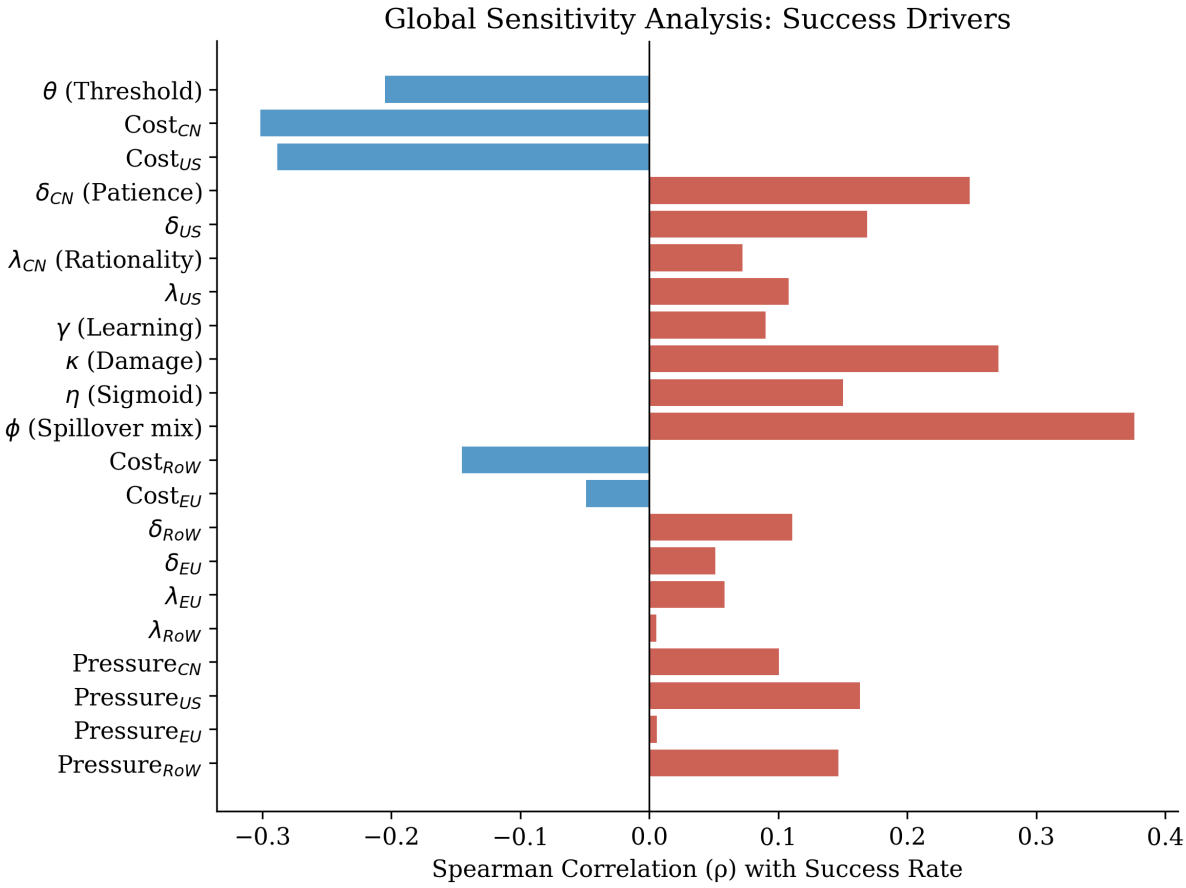
The GSA reports Spearman correlations between each parameter and the two outcomes of interest (coordination success and timing) across 1,000 joint draws.

Policy levers.

Cost_{CN} ($\rho = -0.301$) and Cost_{US} ($\rho = -0.289$) are the largest negative correlates of success, followed by Cost_{RoW} ($\rho = -0.145$); Cost_{CN} has the largest negative correlation with China's first-mover probability of any parameter ($\rho = -0.551$, $p < 0.001$). Cost_{EU} is not statistically significant. Among patience channels, δ_{CN} is the largest positive correlate of success ($\rho = +0.248$) and the largest accelerator of timing overall ($\rho = -0.341$); δ_{US} follows ($\rho = +0.169$); δ_{EU} is not significant. Pressure_{US} ($\rho = +0.163$) and Pressure_{RoW} ($\rho = +0.147$) are positive correlates of success; Pressure_{CN} ($\rho = +0.101$) is marginally significant and Pressure_{EU} is not distinguishable from zero. γ is the smallest significant positive correlate of success ($\rho = +0.090$, $p < 0.01$) with no effect on timing. Among rationality parameters, λ_{US} is negatively correlated with timing ($\rho = -0.223$), λ_{EU} less so ($\rho = -0.128$), and λ_{CN} similarly ($\rho = -0.122$); influence weights are $w_{US} = 0.299$ and $w_{CN} = 0.308$. λ_{US} is negatively correlated with US first-mover probability ($\rho = -0.433$, $p < 0.001$) and positively correlated with China's first-mover probability ($\rho = +0.126$, $p < 0.001$); the EU correlation is marginal ($\rho = +0.054$, $p = 0.09$). λ_{CN} is negatively correlated with China's own first-mover probability ($\rho = -0.159$); cross-bloc correlations are not distinguishable from zero ($\rho = +0.031$, $p = 0.32$ onto the US; $\rho = +0.050$, $p = 0.11$ onto the EU). Appendix C reports this pattern across perturbed cost and patience environments.

Structural parameters.

ϕ is held fixed at 0.5 during SMM calibration and varied across $[0, 1]$ in the GSA; it is the largest positive correlate of coordination success ($\rho = +0.376$, $p < 0.001$) and the largest accelerator of timing ($\rho = -0.411$, $p < 0.001$), ranking first on both dimensions. ϕ and the bloc-specific pressure multipliers are drawn independently; Appendix B reports that the SMM-recovered cardinal weights are stable in the region $\phi \in [0.5, 1.0]$. κ is the second-ranked positive correlate of success ($\rho = +0.271$, $p < 0.001$). θ is a significant negative correlate of success ($\rho = -0.205$, $p < 0.001$). η is negatively correlated with timing ($\rho = -0.100$) and positively correlated with success ($\rho = +0.150$).



Global sensitivity analysis: Spearman ρ between each input parameter and the coordination success rate across 1,000 joint parameter draws.

Table 7: Global Sensitivity Analysis: Spearman Rank Correlations

Parameter	**Success Rate**	**Coord. Period**	**CN First-Mover**	**US First-Mover**
θ (Threshold)	-0.205***	+0.367***	-0.119***	-0.112***
Cost_{CN}	-0.301***	+0.258***	-0.551***	-0.056
Cost_{US}	-0.288***	+0.141***	-0.068*	-0.577***
δ_{CN} (Patience)	+0.248***	-0.341***	+0.488***	+0.033
δ_{US}	+0.169***	-0.188***	+0.045	+0.378***
λ_{CN} (Rationality)	+0.072*	-0.122***	-0.159***	+0.031
λ_{US}	+0.108***	-0.223***	+0.126***	-0.432***
γ (Learning)	+0.090**	+0.035	+0.014	-0.063*
κ (Damage)	+0.271***	-0.087**	+0.125***	+0.068*
η (Sigmoid)	+0.150***	-0.100**	+0.065*	+0.058
ϕ (Spillover mix)	+0.376***	-0.411***	+0.203***	+0.206***
Cost_{RoW}	-0.145***	+0.101**	-0.038	-0.036
Cost_{EU}	-0.049	+0.003	+0.025	+0.005
δ_{RoW}	+0.111***	-0.078*	+0.033	-0.057
δ_{EU}	+0.051	-0.062	-0.022	+0.004
λ_{EU}	+0.058	-0.128***	+0.083**	+0.048
λ_{RoW}	+0.005	-0.038	+0.058	-0.004
Pressure_{CN}	+0.101**	-0.075*	+0.009	+0.068*
Pressure_{US}	+0.163***	-0.125***	+0.081*	+0.136***
Pressure_{EU}	+0.006	-0.015	-0.062*	+0.013
Pressure_{RoW}	+0.147***	-0.167***	+0.081*	+0.077*

Note: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Parameters drawn: $\lambda_i \in [0.1, 5.0]$, $\delta_i \in [0.55, 0.90]$, $\theta \in [0.6, 0.92]$. Negative ρ in Coord. Period indicates faster success.

Table 8: Rationality spillover effects. Spearman ρ between bloc-specific λ and coordination outcomes from 1,000 GSA draws. ** $p < 0.01$, * $p < 0.05$.

Parameter	fm_{US}	fm_{EU}	fm_{CN}	**Coord. timing**
λ_{US}	-0.432**	+0.054	+0.126**	-0.223**
λ_{CN}	+0.031	+0.050	-0.159**	-0.122**

5.5 Interpretation and Identification Scope

The results should be read as identifying a hierarchy of mechanisms rather than providing point estimates of policy effects. The calibration separates ordinal structure, which is disciplined by data, from cardinal scaling, which is recovered from stylised moments; the latter therefore reflects a model-consistent representation of the 2025 strategic environment rather than an empirical estimate of behavioural parameters. Parameters such as the spillover composition ϕ are not point-identified and are instead evaluated over a wide range in the sensitivity analysis, while others (e.g. δ and cost multipliers) map to policy-relevant constructs only at the level of mechanism rather than instrument. The consistency of rankings across marginal sweeps and global sensitivity analysis indicates that the main findings are structural within the model class, but the quantitative crossing thresholds are calibration-dependent and should not be interpreted as precise policy targets.

6 Policy Implications

6.1 Commitment vs Acceleration: Two Distinct Functions

Coordination is driven by two distinct functions. *Commitment drivers* resolve the first-mover problem, moving the system from a state in which no bloc moves to one in which at least one does. *Acceleration drivers* compress the time to coordination once adoption has begun, operating on propagation rather than initiation. Instruments targeting one function do not substitute for those targeting the other, and the policy reading of each lever depends on which role it plays. The subsections below are organised around this split.

6.2 Upfront Costs as the Binding Commitment Constraint

Upfront transition costs determine whether blocs move at all. α_c weights the largely sunk capital burden governments face at the moment of commitment, retiring fossil-fuel capacity, retooling supply chains, and financing replacement infrastructure, rather than the marginal operating cost of clean energy once installed. Because this burden enters the initial decision directly, without routing through any coalition-dependent spillover, cost reductions act on the first-mover problem itself. Learning effects (γ) in principle lower costs too, but only after adoption has occurred, so they cannot resolve the commitment bottleneck, only accelerate it once commitment is underway. Cost reduction therefore triggers coordination while cost learning only accelerates it. The effect concentrates where the burden is largest: Chinese cost is the single strongest determinant of Chinese first-mover probability and a primary determinant of system-wide coordination. The policy mapping corresponds to instruments that directly reduce upfront capital burdens, including concessional finance, technology transfer, and deployment subsidies abroad, and production tax credits, loan guarantees, and procurement-backed deployment domestically.

6.3 Planning Horizons vs Strategic Coherence

Two time-related channels operate through distinct mechanisms. Discount factors δ_i govern how much weight blocs place on future coordination payoffs and therefore enter the willingness-to-commit decision: longer horizons move incentives in favour of early adoption. Rationality parameters λ_i govern how sharply blocs compute best responses and therefore enter the predictability of behaviour: higher λ makes a bloc's strategic behaviour more legible to others. δ moves incentives; λ stabilises expectations.

Patience is not the largest lever in magnitude, but it is structurally robust: the binding individual patience constraint flips with whichever pivotal bloc currently has the shorter horizon, whereas the joint US/China crossing is invariant to that ordering (Appendix D). Moving both horizons together removes the dependency on which bloc happens to be the current laggard, and the joint constraint captures whether the two pivotal players together assign sufficient weight to future payoffs for early adoption to be rational. The rationality asymmetry then operates on the second channel. A more strategically coherent United States is consistently associated with earlier Chinese first-mover adoption, whereas a more coherent China generates no reciprocal effect on US behaviour. As the pivotal endgame mover in the adoption sequence, the US anchors the expectations of blocs that move earlier; reforms that extend and stabilise the US

policy horizon therefore do double duty, improving domestic decision-making and pulling forward adoption elsewhere through the expectations spillover.

6.4 Coalition Formation: Carrot vs Stick

How the coalition grows matters as much as whether it grows. The spillover composition parameter ϕ splits the coalition-growth channel into a carrot (learning and cost-reduction effects that lower the price of joining) and a stick (political and trade-based pressure that raises the price of staying out). The carrot operates on followers, by making accession cheaper for blocs already inclined to move; the stick operates on holdouts, by raising the cost of the outside option directly. The two channels are substitutes in outcome but distinct in target.

The GSA finding is directional rather than magnitudinal, as ϕ is not identified by the moments and is varied across its full range: pressure-dominant environments are consistently associated with faster and more reliable coordination than cost-learning-dominant environments. The implication is that trade-linked pressure instruments, carbon border adjustments, climate-conditioned trade preferences, and diplomatic conditionality, operate as primary levers rather than secondary amplifiers of subsidy-led instruments.

6.5 Structural Conditions Governing Effectiveness

Structure determines whether incentives compound or dissipate. Three parameters describe the strategic environment in which the levers above operate and condition how strongly those levers propagate.

Perceived attainability (θ) is a strategic belief, not a physical participation threshold: whether blocs assign meaningful probability to coordination success given the current coalition state. Its dynamics are endogenous. Early progress raises perceived attainability, which accelerates further adoption, which validates the belief. Institutional architectures that make partial progress legible, staged milestones, early coalition wins, and credible roadmaps, shift this belief without altering underlying physics.

Damage salience (κ) reflects how sharply governments act as if damages are compounding, a function of attribution science, public risk salience, and the credibility of climate forecasts. It pushes pivotal blocs out of the delay equilibrium and raises the baseline probability against which all incentive-moving levers operate. Information-side instruments, national climate risk disclosures, attribution-informed adaptation plans, and sovereign-debt frameworks that price climate exposure, act on this condition.

Threshold shape (η) governs whether cooperation is perceived as collective and binary (high η) or incremental and optional (low η). It has no clean policy mapping and is not an actionable design choice; its role is to calibrate how abrupt the phase transition is once coordination is within reach.

6.6 Interaction Effects: Why Context Matters

Levers do not act in isolation. Cost reductions are more potent under high ϕ, because a pressure-heavy environment raises the payoff to joining relative to staying out; lower perceived thresholds amplify the impact of early coalition wins by converting partial progress into expected success; higher damage salience lifts the baseline probability against which every lever operates. The same policy therefore produces different outcomes depending on the strategic environment, and single-channel strategies are likely to underperform strategies that pair incentive-moving instruments with condition-moving instruments.

6.7 Secondary Channels and Limited Effects

Not all plausible mechanisms are decision-relevant. The learning rate γ is too slow to resolve the commitment problem: cost declines driven by cumulative adoption, though empirically real, arrive after the first-mover decision has already been made, so the main policy route to lower transition costs is direct rather than waiting for learning curves. Per-bloc pressure channels are individually modest once the composition parameter ϕ is included, indicating that the pressure effect is better read through the aggregate composition lever than through any single bloc's trade conditionality in isolation. Sigmoid steepness is structural rather than actionable.

7 Limitations

Three scope conditions bound the analysis. Each points toward a natural extension rather than threatening the core findings.

7.1 RoW Aggregation

The Rest of World bloc pools economies whose strategic incentives diverge in practice: India, Brazil, ASEAN economies, and Gulf states face different cost schedules, trade exposures, and domestic political economies. Aggregating them suppresses heterogeneity that is economically meaningful. The aggregation reflects a binding computational constraint, since the state space grows as $2^n \cdot T$ in the number of blocs n and each additional bloc roughly doubles the nodes requiring fixed-point iteration. The four-bloc specification is the largest system tractable on available hardware without sacrificing the breadth of the sensitivity analysis. The GSA varies RoW cost and discount parameters across wide ranges and confirms the lever hierarchy is not sensitive to RoW parameterisation, but a disaggregated RoW would likely reveal that the cost bottleneck applies unevenly across India, ASEAN, and Latin American economies, and is the most natural extension of the present analysis.

7.2 Irreversibility

The model treats adoption as fully irreversible, which disciplines the strategic problem cleanly but overstates the commitment value of early movers in practice. The US Paris withdrawal illustrates that nominal reversals are politically feasible even if economically costly. A richer specification would allow probabilistic reversal at a cost, introducing a credibility discount on early adoption that would weaken the patience channel findings at the margin. The asymmetry result probably survives since it is driven by the structural weight of US decisions rather than their permanence. Quantitative crossing thresholds would shift under partial reversibility, depending on how reversal costs are parameterised, a scope condition on the quantitative results rather than the qualitative hierarchy.

7.3 Discrete but Non-Binary Commitment

The model restricts each bloc to a binary adoption decision in each period. Real transitions proceed through intermediate stages: partial NDC commitments, sectoral decarbonisation, and graduated policy tightening all represent meaningful strategic moves that binary adoption cannot distinguish from inaction. Discrete commitment levels (say 25%, 50%, 75%, and full adoption) would allow the partial compliance strategies that characterise how China and the US have navigated climate commitments in practice. The cost bottleneck finding for China may be somewhat overstated under binary adoption, since partial adoption available at lower cost would lower the threshold for Chinese movement. Whether graduation dynamics materially alter the lever hierarchy remains open.

7.4 Robustness summary

These scope conditions are real but bounded. The three primary findings, the joint patience lever, China's cost bottleneck, and the rationality asymmetry, survive the spillover composition sensitivity check in Appendix B, hold across the perturbed cost and patience environments in Appendix C, are not artefacts of the baseline discount ordering as confirmed in Appendix D (including the US/CN flip), and sit on a calibrated parameter vector confirmed as a unique equilibrium in Appendix A. The qualitative hierarchy is not sensitive to how the spillover composition is mixed, to perturbations of the calibrated cost and patience environment, or to which bloc is assigned the lowest starting discount factor. What remains open is quantitative: the exact crossing thresholds shift under alternative assumptions, but the ordering of levers does not.

8 Conclusion

Climate coordination in this paper is framed as a dynamic threshold problem in which timing, expectations, and asymmetry jointly determine outcomes. The model is not designed to predict specific policy paths, but to identify which mechanisms move the system from delay to coordination and how those mechanisms interact.

The central result is that coordination is governed by two distinct functions. Commitment drivers determine whether blocs move at all, while acceleration drivers shape how quickly coordination is reached once adoption begins. Upfront transition costs emerge as the primary commitment constraint: they enter directly into the first-mover decision and therefore determine whether coordination is initiated. By contrast, mechanisms such as cost learning operate only after adoption has begun and cannot resolve the initial bottleneck. Coalition formation dynamics then shape the path that follows. The composition of spillovers matters: environments in which non-participation is costly tend to generate faster and more reliable coordination than those that rely on lowering entry costs alone.

Time-related channels operate through a different margin. Planning horizons affect the willingness to commit, while strategic coherence affects how expectations are formed and transmitted across blocs. The asymmetry between pivotal players is particularly important: when the endgame mover behaves in a predictable way, it anchors expectations and pulls forward adoption elsewhere, even if its own first-mover incentives remain weak.

These levers operate within a strategic environment that conditions their effectiveness. Perceived attainability of coordination, the salience of future damages, and the degree to which benefits are seen as collective all shape whether incentives compound or dissipate. Policy therefore acts not only by shifting payoffs directly, but by altering the conditions under which those payoffs are evaluated. The same intervention can produce different outcomes depending on how credible, urgent, and attainable coordination is perceived to be.

The analysis is subject to clear limitations. The model abstracts from country-level heterogeneity beyond four blocs, relies on stylised calibration targets, and treats key parameters, particularly spillover composition, as exogenous to policy formation. These simplifications are deliberate, allowing the identification of mechanisms rather than precise empirical prediction.

Taken together, the results suggest that climate coordination failure is not primarily a problem of technological feasibility, but of strategic timing under uncertainty. The key challenge is not whether coordination is possible, but how to move the system into a region where commitment occurs early enough for acceleration mechanisms to take effect.

A Appendices

A.1 Equilibrium Uniqueness

QRE is defined as a fixed point in the mapping from beliefs to logit best-response probabilities. If multiple fixed points exist, findings depend on equilibrium selection rather than the underlying strategic structure. Two tests rule this out at the SMM baseline.

The fixed-point iteration uses damping: $\sigma_{k+1} = \alpha F(\sigma_k) + (1-\alpha)\sigma_k$ with $\alpha = 0.5$, where F is the logit best-response map. Damping preserves the fixed points of F but improves convergence behaviour when the undamped iteration is slow or non-contractive. At the SMM-calibrated vector, F is already a contraction ($\rho(J_F) = 0.804$ at its worst node), so damping is not strictly necessary at the baseline. It is retained because across the GSA parameter draws the undamped map can push above one at isolated nodes, and damping ensures that every draw in the sensitivity analysis converges to its equilibrium.

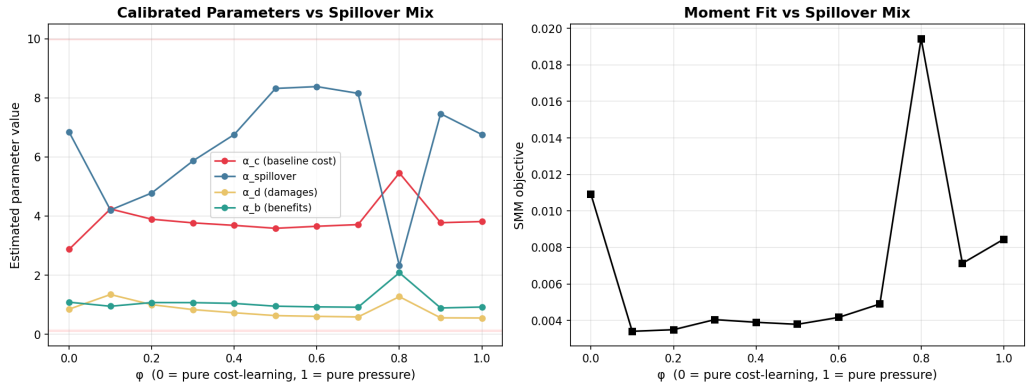
Both tests are conducted at the SMM-calibrated parameter vector, checking each of the 110 game-tree nodes with two or more active players. The first test initialises the fixed-point iteration from 20 random starting points plus simplex corners at every node. All initialisations converge to identical equilibrium probability vectors, with a maximum deviation of 6.70×10^{-8}; no node exhibits disagreement above the 10^{-5} tolerance. The second test computes the spectral radius of the damped best-response Jacobian $J_T = \alpha J_F + (1-\alpha)I$ at the converged solution. A spectral radius below one confirms the fixed point is a strict local contraction. The maximum spectral radius across all nodes is $\rho(J_T) = 0.861$, with a median of 0.529 and 99th percentile of 0.784. Together the two tests establish that the equilibrium characterised throughout the analysis is unique at every game-tree node at the baseline, and the damped solver ensures the same guarantee extends to every parameter draw visited by the sensitivity analysis.

Table 9: QRE Equilibrium Uniqueness Tests at SMM-calibrated parameters. Tests 1–2: fixed-point re-initialised from 20 random starts plus simplex corners at every active game-tree node. Test 3: spectral radius of the logit best-response Jacobian J_F at σ^*.

Test	**Criterion**	**Result**	**Verdict**
Multi-start (Tests 1–2)	Max deviation from σ^* across all nodes	7.52e-08	Pass
	Non-converged initialisations	0	
	Active nodes tested	150	
Eigenvalue stability (Test 3)	Max spectral radius $\rho(J_F)$ (need < 1)	0.8842	Pass
	Nodes with ≥ 2 active players	110	

A.2 SMM Sensitivity to ϕ

Cost-learning and political pressure both intensify with cumulative adoption W_t and generate observationally equivalent adoption paths, so no moment constructed from trajectories can cleanly separate their cardinal weights. This is the identification problem that motivated collapsing the two mechanisms into a single composite spillover channel scaled by α_{spill}, with ϕ controlling the internal cost-learning/pressure split. Because ϕ is unidentified from the moments, it is held fixed at 0.5 during SMM estimation. This appendix asks whether the four cardinal weights $(\alpha_c, \alpha_{\text{spill}}, \alpha_d, \alpha_b)$ depend on that choice. The full four-parameter SMM is re-estimated at each $\phi \in \{0.0, 0.1, \ldots, 1.0\}$ and the recovered trajectories are plotted in Figure 7.



SMM-calibrated parameters as a function of the spillover mixing parameter ϕ. The rank ordering $\alpha_{\text{spill}} > \alpha_c \gg \alpha_b > \alpha_d$ is preserved at every converged point. The spike at $\phi = 0.8$ is a failed local optimisation (SMM objective ≈ 0.019 against ≈ 0.004–0.009 at neighbouring points) rather than a genuine parameter shift.

The four trajectories are broadly stable but not flat, and the shape of the α_{spill} curve is worth dwelling on. From $\phi = 0.1$ upward the spillover weight climbs almost monotonically — $4.20 \to 4.78 \to 5.87 \to 6.76 \to 8.32 \to 8.38$ at $\phi = 0.6$ — then eases back to 8.15 at $\phi = 0.7$, collapses to 2.32 at $\phi = 0.8$, and recovers to 7.46 and 6.75 at $\phi = 0.9$ and $\phi = 1.0$. The hump shape is what identification theory would predict: as ϕ moves from "pure cost-learning" toward a balanced mix, a larger α_{spill} is needed to push the same cumulative W_t through both channels simultaneously and reproduce the observed adoption moments; beyond $\phi \approx 0.6$–0.7 the mechanics tip toward political pressure, which feeds into payoffs more efficiently and lets α_{spill} ease back. The collapse at $\phi = 0.8$ does not fit this logic — the objective there is an order of magnitude worse than at $\phi \in \{0.7, 0.9\}$ and the recovered vector ($\alpha_c = 5.46$, $\alpha_{\text{spill}} = 2.32$, $\alpha_b = 2.08$) sits far off the smooth trajectory traced by adjacent points, so it reads as a failed local optimisation rather than a real discontinuity. Flagging rather than patching keeps the estimator's behaviour visible.

The remaining three weights move within narrower envelopes. Across $\phi \in [0.5, 1.0]$ (five clean points) $\alpha_c \in [3.58, 3.81]$, $\alpha_d \in [0.55, 0.63]$, and $\alpha_b \in [0.89, 0.95]$; across the full unit interval the spreads widen to $\alpha_c \in [2.88, 4.24]$, $\alpha_d \in [0.55, 1.34]$, and $\alpha_b \in [0.89, 1.08]$, driven mostly by the near-pure cost-learning endpoints ($\phi \in \{0.0, 0.1\}$).

This stability matters for the interpretation of the main results. The policy hierarchy reported in Section 6 is built on top of the cardinal vector recovered by SMM: the channel importance rankings depend on the relative magnitudes of α_c, α_{spill}, α_d, and α_b. If the rank ordering flipped when ϕ was reweighted, the

channel rankings would be conditional on the particular ϕ assumption fed into estimation, and the main findings would be a feature of one specific spec rather than a structural property of the model. Figure 7 rules that out: the ordering $\alpha_{\text{spill}} > \alpha_c \gg \alpha_b > \alpha_d$ is preserved at every converged point, so the policy ordering inherited from the calibration does not depend on which cost-learning/pressure split is fixed at estimation.

A.3 Robustness of the λ_{US} Cascade

The rationality discussion in Section 6 rests on a specific claim: a more rational United States pulls back from period-one adoption but triggers a positive first-mover cascade onto the EU and China, accelerating overall coordination. If this result only holds at the precise baseline calibration, it is a feature of one parameter point rather than a structural property of the game. This appendix verifies the cascade across perturbed cost structures and discount factors.

At each of nine grid points, the cost scaling parameter α_c is multiplied by $\{0.75, 1.00, 1.25\}$ and all four bloc-level discount factors are shifted by $\{-0.05, 0.00, +0.05\}$, producing a 3×3 grid of perturbed environments. At each grid point λ_{US} is swept from 0.5 to 5.0 across the full experimental range, and the change in period-one adoption probabilities and mean coordination timing is recorded. The cascade is defined to hold if all four diagnostic signs are present: $\Delta\sigma_{US}(1) < 0$ (US pulls back), $\Delta\sigma_{EU}(1) > 0$ (EU cascades), $\Delta\sigma_{CN}(1) > 0$ (CN cascades), and Δ timing < 0 (coordination arrives earlier).

Table 10: Robustness of the λ_{US} cascade across a 3×3 grid of perturbed cost (α_c) and patience (δ) environments. At each grid point, λ_{US} is swept from 0.5 to 5.0 with other blocs' λ held at the SMM baseline; reported values are the high-minus-low differences. The cascade holds if all four diagnostics take the expected sign ($\Delta\sigma_{US}(1) < 0$, $\Delta\sigma_{EU}(1) > 0$, $\Delta\sigma_{CN}(1) > 0$, Δ timing < 0). Grid points where coordination collapses entirely at both endpoints are marked "collapse"; the cascade is vacuously undefined there rather than violated.

α_c scale	δ shift	$\Delta\sigma_{US}(1)$	$\Delta\sigma_{EU}(1)$	$\Delta\sigma_{CN}(1)$	Δ timing	Success (lo / hi)	Verdict
0.75	-0.05	-0.223	+0.113	+0.195	-0.37	0.91 / 0.93	✓
0.75	+0.00	-0.226	+0.077	+0.394	-0.75	0.98 / 1.00	✓
0.75	+0.05	-0.207	+0.040	+0.388	-0.82	0.99 / 1.00	✓
1.00	-0.05	-0.164	-0.018	+0.002	-0.25	0.63 / 0.03	×
1.00	+0.00	-0.196	+0.201	+0.240	-0.51	0.77 / 0.92	✓
1.00	+0.05	-0.206	+0.084	+0.458	-0.96	0.96 / 1.00	✓
1.25	-0.05	-0.103	-0.006	-0.000	–	0.21 / 0.00	collapse
1.25	+0.00	-0.142	-0.018	+0.001	+0.45	0.49 / 0.01	×
1.25	+0.05	-0.186	+0.260	+0.424	-0.69	0.71 / 0.97	✓

The cascade pattern holds at six of nine grid points. The three non-holds all sit at α_c scalings of 1.00 or 1.25 combined with non-positive discount shifts, and all share the same underlying failure mode: coordination success collapses in the high-λ_{US} regime (success rate below 0.05 at $\lambda_{US} = 5.0$), so the timing comparison between the endpoints is vacuous rather than substantively contradicted. When the coalition fails to form, the timing measure carries no information, and the $\Delta\sigma_{EU}(1)$ and $\Delta\sigma_{CN}(1)$ diagnostics contract toward zero because neither bloc responds strongly to a US anchoring to a collapsed state. Across every region of parameter space where coordination remains feasible, increasing λ_{US} reduces US period-one adoption while simultaneously raising others' and accelerating timing. The cascade is therefore structural: it reflects the United States' position as the pivotal endgame mover whose credible patience anchors others' best-response calculations, not a local artefact of the SMM-calibrated parameter vector.

A.4 Discount Ordering Robustness

The bloc-level discount factors in the baseline (EU 0.85 ¿ CN 0.80 ¿ US 0.75 ¿ RoW 0.70) are anchored to a narrative argument about institutional credibility rather than to measured time preferences.[1] Two concerns follow. The first is that the patience-dominates finding could be an artefact of the specific *ordering*: if the US were assigned higher patience than China, the binding constraint would shift, and US patience would no longer appear as the strongest individual lever. The second is that the uniform 0.05 per-bloc *spacing* between horizons is also a convenience choice: if the bloc horizons were compressed into a tighter range or expanded into a wider one, the joint patience channel might lose its leading position. This appendix tests both concerns across eight alternative configurations, including the US/CN flip, a fully equalised vector, US-most-patient and China-most-patient orderings, and three spacing variants (compressed to 0.78/0.82, compressed to uniform 0.02 gaps, and expanded to uniform 0.08 gaps). At each configuration the SMM-calibrated cardinal vector is held fixed and only the discount factors change; sweep crossings are reported as percentage shifts from each configuration's own baseline rather than a hardcoded global default.

Table 11: Robustness of the patience-dominates finding across eight alternative discount-factor configurations. The SMM-calibrated cardinal vector $(\alpha_c, \alpha_d, \alpha_p, \alpha_b, \lambda)$ is held fixed; only the per-bloc discount factors change. Each entry is the percentage shift from that configuration's own baseline required to reach a 90% coordination success rate on the given sweep channel. "absent" indicates the 90% threshold is not reached within the sweep range. Grid: 14 points, 250 Monte Carlo draws per point.

Configuration	P_0	δ_{US}=δ_{CN}	Global cost	δ_{US}	δ_{CN}	δ_{EU}	θ
Baseline (EU¿CN¿US¿RoW)	0.76	+4.7	-8.5	+8.2	+5.8	+7.2	-24.5
US/CN flipped	0.66	+4.7	-16.1	+10.1	+8.2	+11.8	absent
All equalised (0.775)	0.66	+4.7	-16.1	+13.7	+9.2	+15.1	-24.5
US most patient	0.80	+1.4	-8.5	+3.6	+8.2	+6.7	-11.5
CN most patient	0.97	+1.4	+6.9	-5.6	-0.5	-12.5	+5.8
Compressed 0.78/0.82	0.89	+1.4	-0.8	+2.3	+0.6	-5.3	+1.4
Uniform 0.02 gaps	0.82	+2.7	-8.5	+8.5	+5.8	+6.5	-11.5
Uniform 0.08 gaps	0.17	+9.7	-16.1	+20.9	+12.9	absent	-37.5

The joint US/China patience channel reaches the 90% coordination frontier across every configuration tested. In the challenging environments (baseline, US/CN flipped, all-equal, and uniform 0.02 gaps) the joint channel crosses with shifts of 2.7% to 4.7% while global cost reduction requires 8.5% to 16.1%. In the three permissive configurations (US-most-patient, China-most-patient, compressed 0.78/0.82) the baseline success rate already sits at or above 0.80 and the system either reaches 90% with a small additional shift or is already there. The expanded uniform-0.08 spacing is the most demanding environment by a wide margin: pulling the bloc horizons apart drops the US to 0.70 and RoW to 0.62, and the baseline success rate collapses to 0.17. Even here the joint channel reaches 90% at +9.7% while global cost reduction needs −16.1%, and the individual δ_{EU} channel is absent entirely. The joint channel is the cheapest 90% lever in every case.

The US/CN flip is the most informative single ordering test. Swapping the discount factors so the US is the more patient of the two pivotal blocs (0.80 vs 0.75) shifts the individual-lever numbers (δ_{US} now

[1] The crossing values reported in this appendix are computed on a coarser sweep grid (14 points, 250 Monte Carlo draws per point) than the main marginal sweeps in Section 5.2 (denser grid, $n = 1{,}000$ MC draws). The joint US/China crossing therefore appears here as +4.7% rather than the +6.75% reported in Table 6; both values describe the same underlying fixed point, and the difference reflects the resolution at which the success-rate curve is interpolated. The qualitative ordering of levers and the cross-configuration comparisons within this appendix are unaffected.

requires a 10.1% shift and δ_{CN} an 8.2% shift, versus 8.2% and 5.8% respectively in the baseline), but leaves the joint channel crossing at the same +4.7% in either ordering. The two uniform-spacing tests extend the same conclusion to the range axis: a 0.02-gap vector preserves the hierarchy at a smaller scale, and a 0.08-gap vector preserves it at a much larger scale even as the system is otherwise close to total coordination failure. The right reading of the patience-dominates result is therefore not ordering-specific and not spacing-specific: the joint channel's efficiency is structural, and its lead over global cost reduction and every individual δ holds regardless of which bloc is currently lagging or how tightly the bloc horizons are spaced.

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